ANGAD SHAH

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PROFESSIONAL SUMMARY

Final-year Computer Science student and aspiring Software Development Engineer (SDE) with proven experience designing and delivering end-to-end full-stack, IoT, and AI-driven applications. Proficient in Python, React, FastAPI, and PostgreSQL with hands-on experience in REST API design, machine learning model integration, cloud deployment on AWS, Docker containerization, and CI/CD pipelines using GitHub Actions. Passionate about building scalable, production-ready backend systems and solving complex engineering problems through clean, maintainable code.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), JavaScript, C++, Java (Familiar)

Frontend: React.js, Next.js, HTML5, CSS3

Backend: FastAPI, Node.js, REST API Design, System Design, Microservices

AI / ML: Pandas, Scikit-learn, Data Analysis, Feature Engineering, Model Building

Databases: PostgreSQL, MongoDB

Cloud & DevOps: AWS (EC2, S3, Lambda, Serverless), Docker, GitHub Actions (CI/CD), Linux, Bash

IoT & Embedded: ESP32, LoRa Wireless Communication, Sensor Integration

Tools & Practices: Git, Postman, API Integration, Agile, Unit Testing, Debugging

PROJECT EXPERIENCE

IoT Predictive Maintenance System 2.0

Nov 2025 – May 2026

github.com/Angadrgg/Predictive-Maintenance-System-2.0

Python | FastAPI | React | AWS Lambda/S3 | ESP32 | LoRa | Scikit-learn | Docker

- Architected an end-to-end IoT predictive maintenance platform integrating ESP32 microcontrollers and LoRa wireless communication for real-time industrial motor health monitoring.
- Designed a serverless AWS architecture using Lambda and S3 to ingest, store, and process sensor data at scale, eliminating infrastructure management overhead.
- Built a machine learning pipeline with Scikit-learn and Pandas for anomaly detection on vibration, temperature, and current sensor streams, achieving high-confidence fault prediction.
- Developed a React dashboard to visualize live sensor telemetry, predictive alerts, and motor health trends with real-time WebSocket data updates.
- Containerized all backend microservices with Docker and automated deployment through GitHub Actions CI/CD, enabling zero-downtime releases.

IoT Predictive Maintenance System 1.0

Nov 2025 – Dec 2025

github.com/Angadrgg/Predictive-Maintenance-System-1.0

Python | FastAPI | React | PostgreSQL | Scikit-learn | AWS EC2/S3 | Docker

- Engineered an end-to-end predictive maintenance system using Scikit-learn, achieving ~85% accuracy in real-time equipment anomaly detection across 10+ sensor channels.
- Architected and deployed a RESTful API layer using FastAPI to ingest, validate, and serve live sensor data with sub-100ms response times.
- Designed an interactive React dashboard to visualize system health metrics, live alerts, and model prediction outputs with dynamic state management.

- Automated data preprocessing and feature engineering using Pandas, standardizing multi-channel sensor inputs for downstream ML model consumption.
- Deployed backend services on AWS EC2 with S3 for persistent model artifact storage; containerized with Docker for reproducible builds.

FitFusion — AI-Powered Fitness Coach May 2026

github.com/Angadrgg/fit-fusion

JavaScript | React | Node.js | REST APIs | AI/ML Integration

- Built a full-stack AI fitness coaching application that generates personalised workout plans and nutrition guidance based on user goals and fitness level.
- Integrated AI model APIs to dynamically generate adaptive training recommendations, progress feedback, and injury-prevention tips.
- Designed a responsive React frontend with an intuitive onboarding flow, progress tracking dashboards, and animated workout visualisations.
- Implemented a Node.js REST API backend to manage user profiles, session history, and AI-generated plan storage with secure authentication.
- Applied RESTful best practices with structured error handling, input validation, and rate limiting to ensure a reliable, production-grade backend.

EDUCATION

Bachelor of Engineering in Computer Science Expected June 2026

JIS College of Engineering | Kalyani, West Bengal | CGPA: 8.46 / 10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, System Design

CERTIFICATIONS

AWS Academy Graduate — AWS Academy Cloud Foundations — Amazon Web Services 2025

The Complete Python Bootcamp: From Zero to Hero in Python — Udemy 2024

Euphoria GenX — Full Stack Development Program — Euphoria GenX 2024